RoomCleaner

A four-cable ceiling robot that picks laundry off the floor

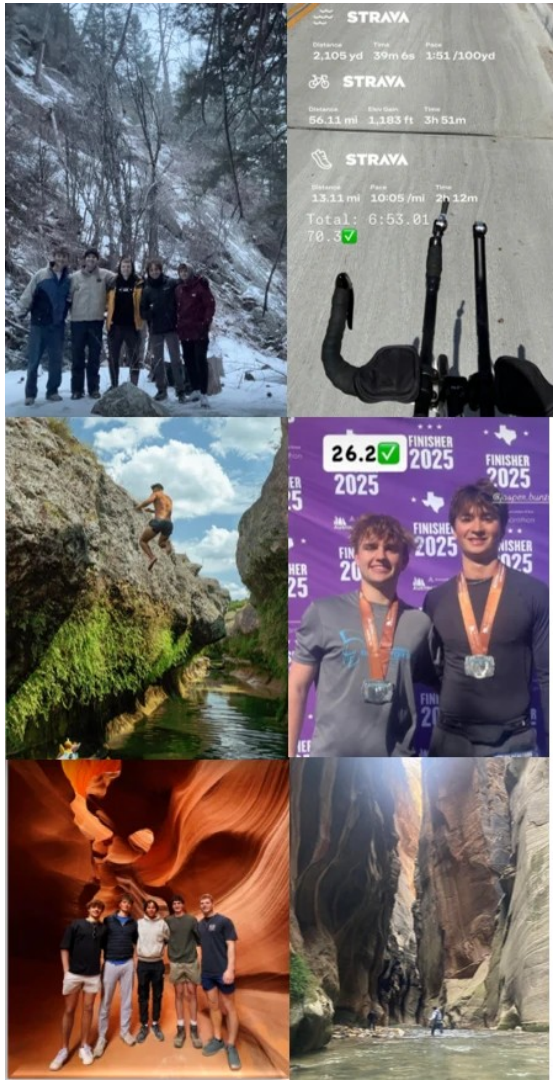
Jasper Buntinx

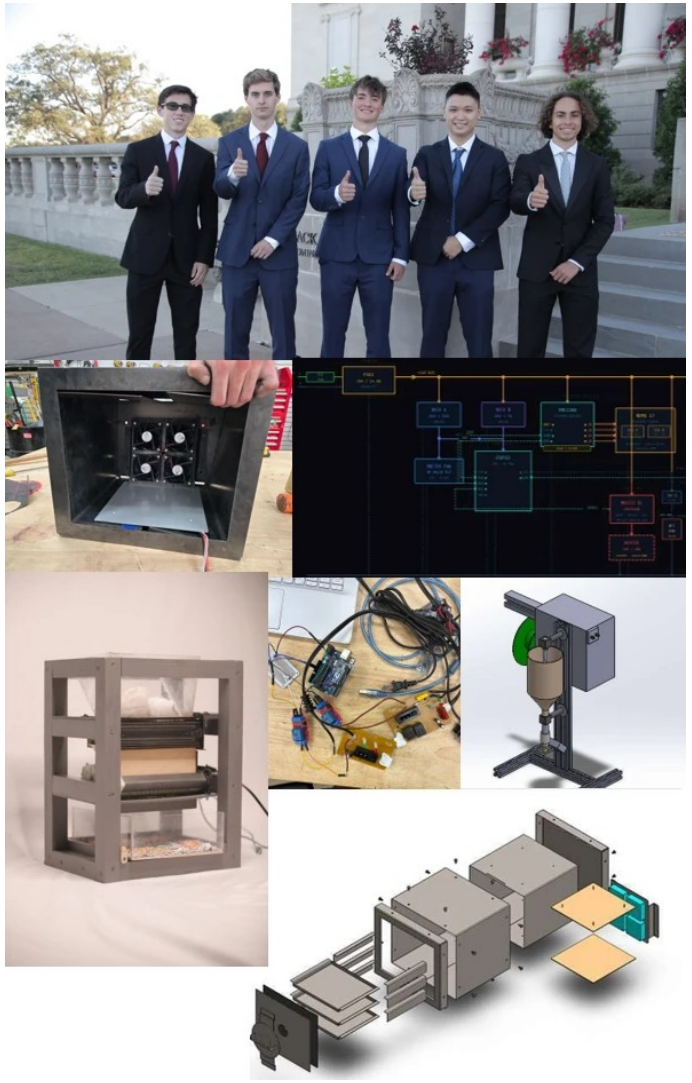
Mechanical Engineering, Texas A&M University

Built July to October 2026



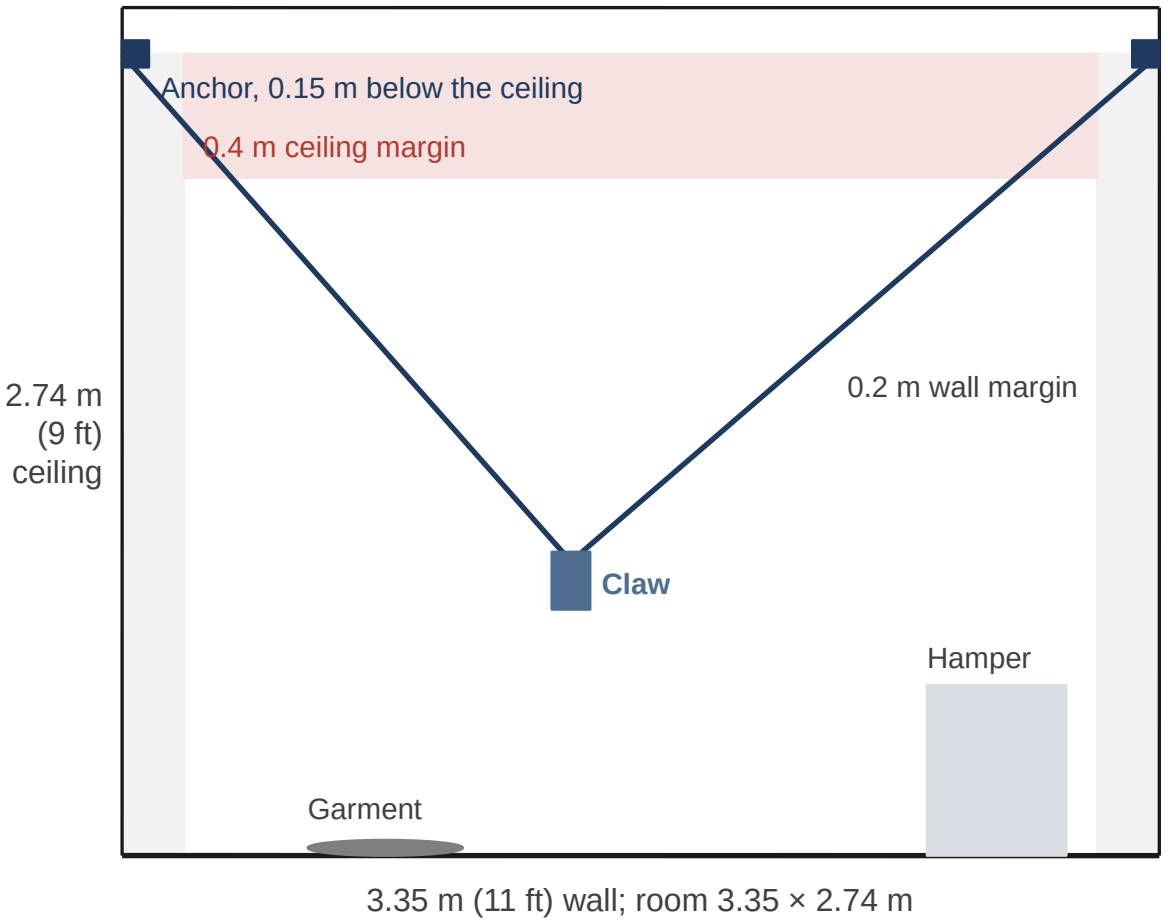
About me


STRAVA
2,105 yd 39m 6s 1:51 /100yd
STRAVA
56.11 mi 1,183 ft 3h 51m
STRAVA
13.11 mi 10:05 /mi 2h 12m
Total: 6:53.01
70.3
26.2
FINISHER 2025




ONE ARMY
GLADIATOR DASH
$185K
RAISED FOR
STILL CREEK RANCH
FINISH

The goal: jeans from anywhere on the floor into the hamper, nobody touching it

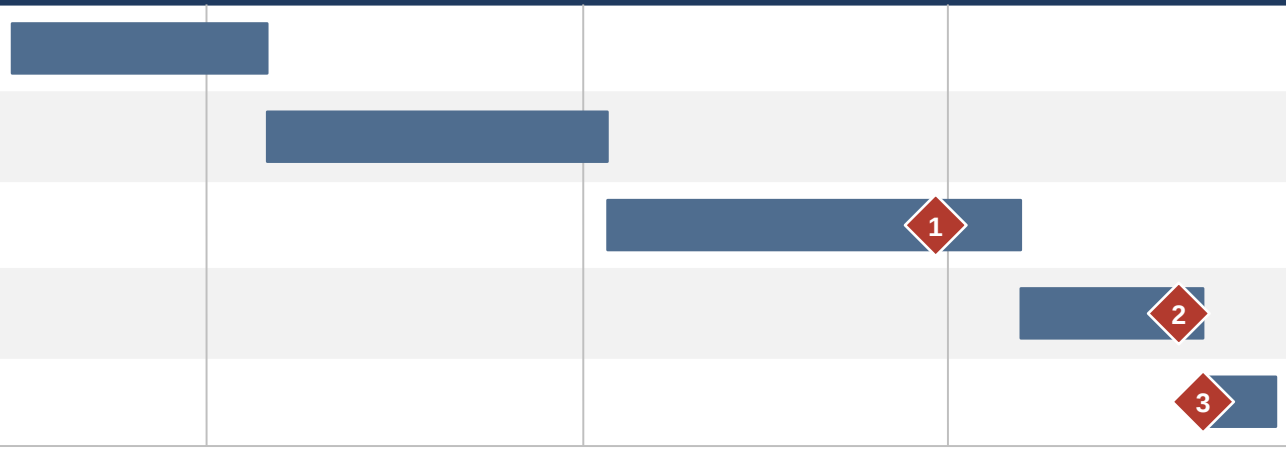


Room elevation to scale; seen from the side, the four cables appear as two

- Pick up laundry in a 3.35 × 2.74 m (11 × 9 ft) room under a 2.74 m (9 ft) ceiling; design load 0.9 kg (jeans) plus a 2 m/s² dynamic allowance
- Four winches screwed into the studs near the ceiling corners, one claw on four cables, nothing on the floor
- PAUSE holds position with torque on; the switched power strip is the emergency stop, and it drops the claw
- Success: one garment from floor to hamper with no human touch; I approve each move on the phone

Five phases in fifteen weeks

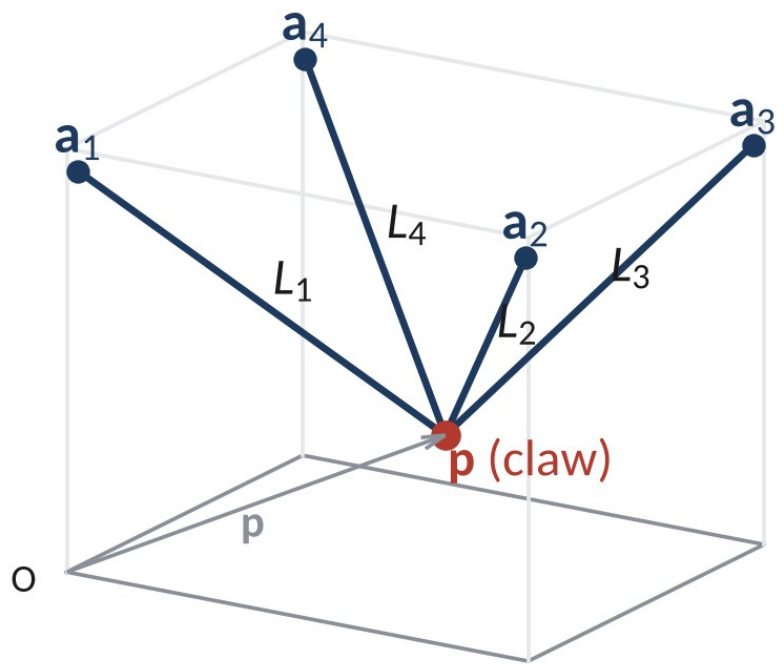
Act	Dates	July	August	September	October
1. Concept	Jul 16 – Aug 5				
2. Generation 1	Aug 6 – Sep 2				
3. Redesign	Sep 3 – Oct 6				
4. Final design	Oct 7 – Oct 21				
5. Build and test	Oct 22 – Oct 27				



1 Sep 30: first clamp's jaws cannot close on thin cloth 2 Oct 20: v6 clamp is 4.6 times its mass budget in CAD 3 Oct 22: drivers at 1.5 A; claw weighed

- Mine: requirements, design decisions, the SolidWorks station and claw, printing, machining, wiring and bring-up
- With Claude Code, under my direction: the control software, calculation scripts and CAD checks
- Purchased: steppers, drivers, controller boards, servo, camera, battery and encoder boards
- Self-imposed limits: a hobby budget (about $640 spent, $220–260 of it in the robot), a 450 g claw, 60 °C for loaded printed parts, and a build week that ends on presentation day

Why a cable robot: four winches cover the whole floor with nothing in the room



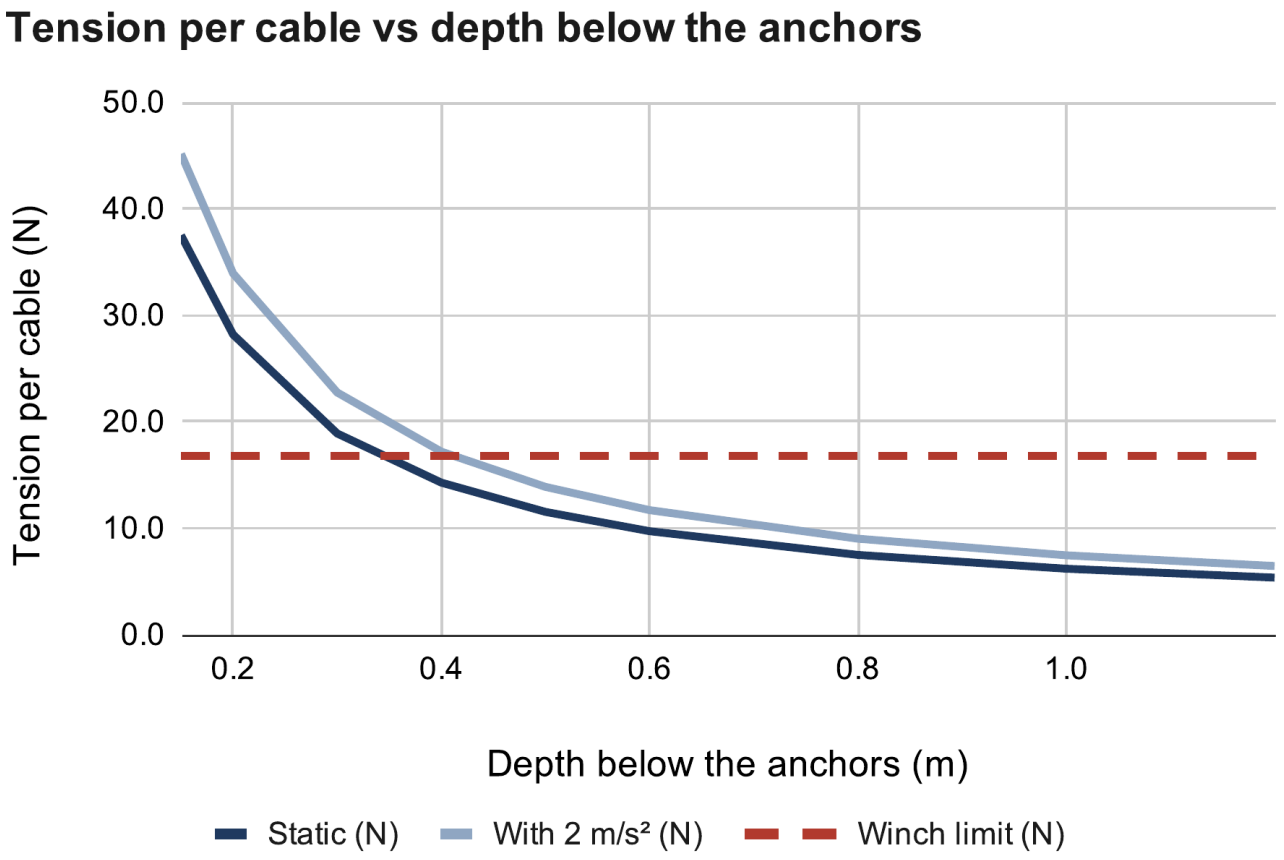
	Floor robot	Gantry	Cable robot
Whole floor	Furniture blocks it	Yes	Yes
Floor left clear	No	Frame spans the room	Yes
Moving mass	Whole robot	Heavy beam	0.6 kg claw

- Chosen: four ceiling winches and one claw
- The cost: cables only pull, so cable tension sets the working space

Cables from the ceiling anchors a_1 to a_4 to the claw at p

The first physics: cable tension climbs toward the ceiling

CALCULATED



Calculated (calc sheet: Cable tension)

- Weighed claw plus 1 lb, centred 0.4 m below the anchors: 14.3 N static, 17.2 N at 2 m/s²
- That is at the 16.8 N winch limit; with jeans the claw must stay at least 0.6 m below the anchors
- The planner checks every cable at every path point against the measured pull limit

Tension per cable, 1 lb test at 0.4 m

$T = m\,(g + a) \,/\, (4 \sin \varphi)$

$= 1.054 \times 11.81 \,/\, (4 \times 0.181) =$ **17.2 N**

Winch limit: **16.8 N**

July's choices, and what each one cost later

WINCH MOTOR

NEMA 17 steppers

CHOSEN BECAUSE

- Cheap, and one shield drives all four
- Holds position without a feedback loop

WHAT IT COST LATER

- No brake: it holds only while powered, so a power loss drops the claw
- Runs hot: holding current is heat, 4.05 W per motor at 1.5 A

GRIPPER

Five tendon fingers

CHOSEN BECAUSE

- Gathers crumpled cloth with one servo
- Printable in flexible TPU

WHAT IT COST LATER

- Designed in CAD only, 419 g of a 450 g budget, never built
- Replaced by a clamp, then by the geared four-bar (Act 3)

EMERGENCY STOP

The power strip

CHOSEN BECAUSE

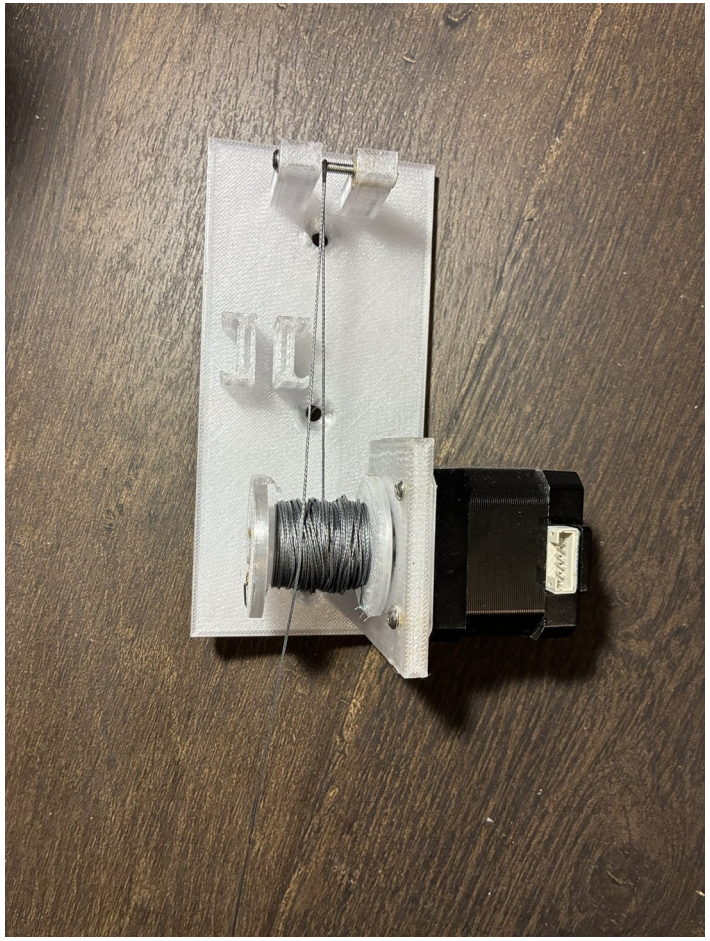
- A guaranteed stop with no extra part
- The only stop that cannot be overridden by software

WHAT IT COST LATER

- Cutting power drops the claw
- Nobody stands under the robot; a brake is on the next-steps list

Generation 1: one Uno, one shield, four corner mounts

MEASURED



1 Gen-1 corner plate: motor, spool, pulley, switch

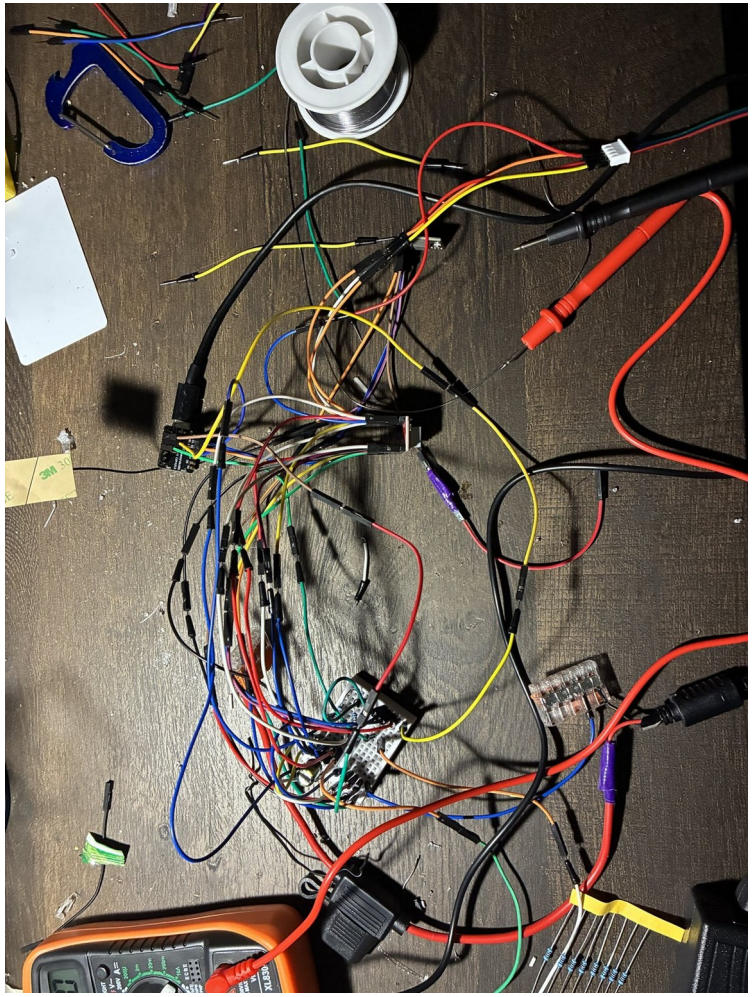


2 Gen-1 station lifting a test weight

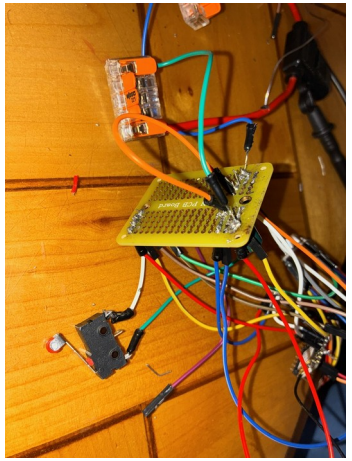
- A printed corner plate screwed into a joist replaced hanging hooks: a hook is a pivot, and about 1:1 of anchor error becomes claw position error
- One Uno and one CNC shield drove all four motors from one board
- Bench: the station lifted a test weight; the cable math held

Bench week found what the drawings could not

MEASURED



1 Bench wiring during bring-up



2 Homing switch

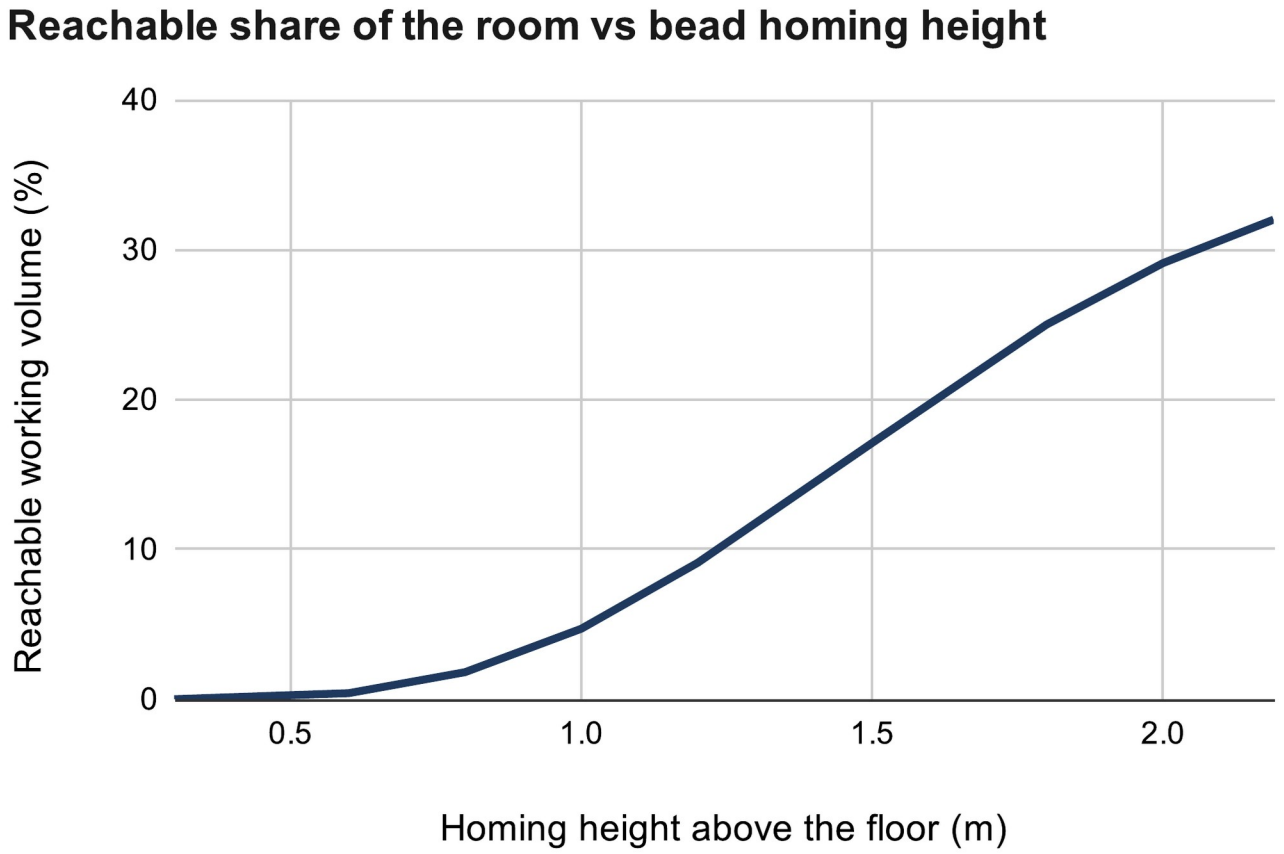


3 Bench laptop

- 18 V on the meter from a supply labelled 12 V
- A homing switch that would have zeroed without moving: fail-safe wiring reads LOW at rest, and the firmware took LOW as pressed; logic inverted
- The switch had to move below the pulley, onto a drop arm beside the line; the first arm design failed a CAD check before it was printed

Why Generation 1 could not be the final design

CALCULATED

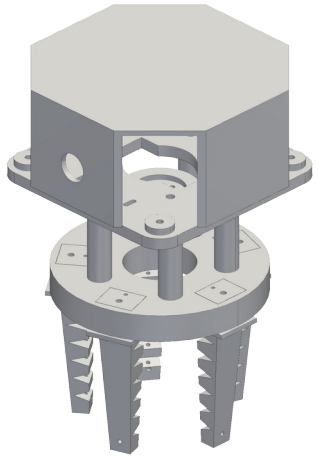


- Bead homing (a stopper bead on each cable trips the outlet switch, so no cable can ever be shorter than at home) left 5–29 % of the room's volume reachable: home 1 m up, 5 %; 2 m up, 29 %. Caught on paper
- Floor area at grab height reads better, 45 % with home 1 m up and 93 % at 1.5 m: what bead homing loses is height, for the scan and the hamper, not footprint
- One central board would need 10 wires to every corner once each winch had sensors: 250 m (820 ft) of wire
- A central board also skews the winches: 50 ms of skew is 2.5 mm of cable error, about 7.5 N

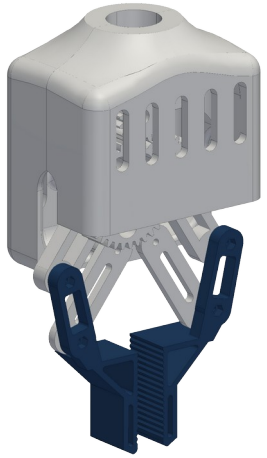
Calculated (calc sheet: Bead homing)

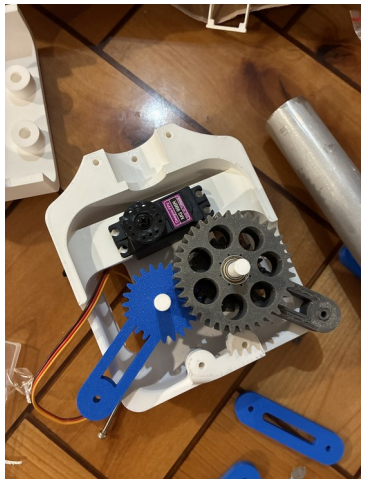
The clamp that could not close on thin cloth (Sep 30)

CAD







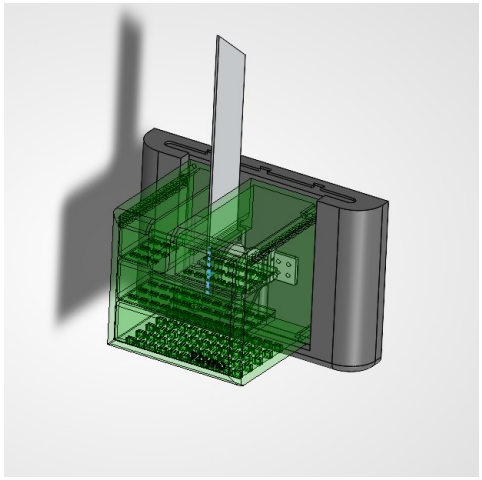


Finger claw (CAD only), v6 clamp (CAD), final four-bar (CAD), printed gear train

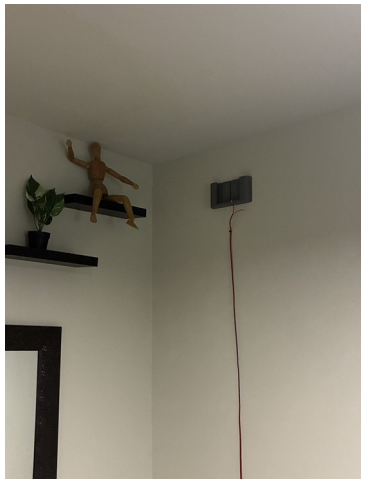
- Rack-and-pinion jaws could not pinch thin fabric: the pinion travel left a 2 mm gap, so the pads never met
- The v6 clamp came out at 4.6 times its mass budget in CAD (Oct 20), with no torque path from the servo
- Lesson: mass tracked through every version from then on; the final claw was weighed on Oct 22
- Final: a geared four-bar with parallel jaws (Act 4)

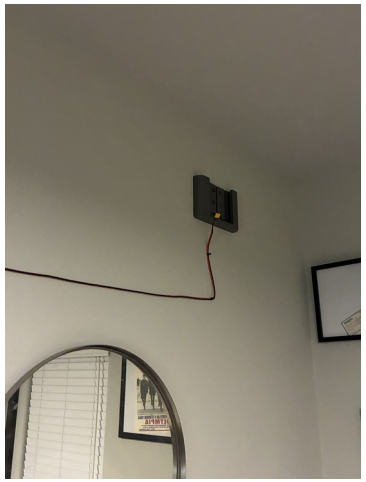
One plug per station: the station lifts off while the anchor stays in the stud

CALCULATED









SolidWorks, shell transparent; the shell on the wall (Oct 22); the wall mount alone in a corner, and with its XT60 power lead (Oct 27)

- Slide-in wall mount: the station lifts off its dovetail; the wall mount and its screws stay in the stud
- Load path: cable, spool, set screw, motor, PETG tab, slide-in, wall mount, screws, stud
- The motor is the weakest link: it slips at 34–59 N before anything breaks, and the spool encoder reports the overload
- Reading the margins: every other part must give at 1.5 × the motor's slip force (calculated, against derated PETG); the 1.5 on the winch pull is the skipped-step margin

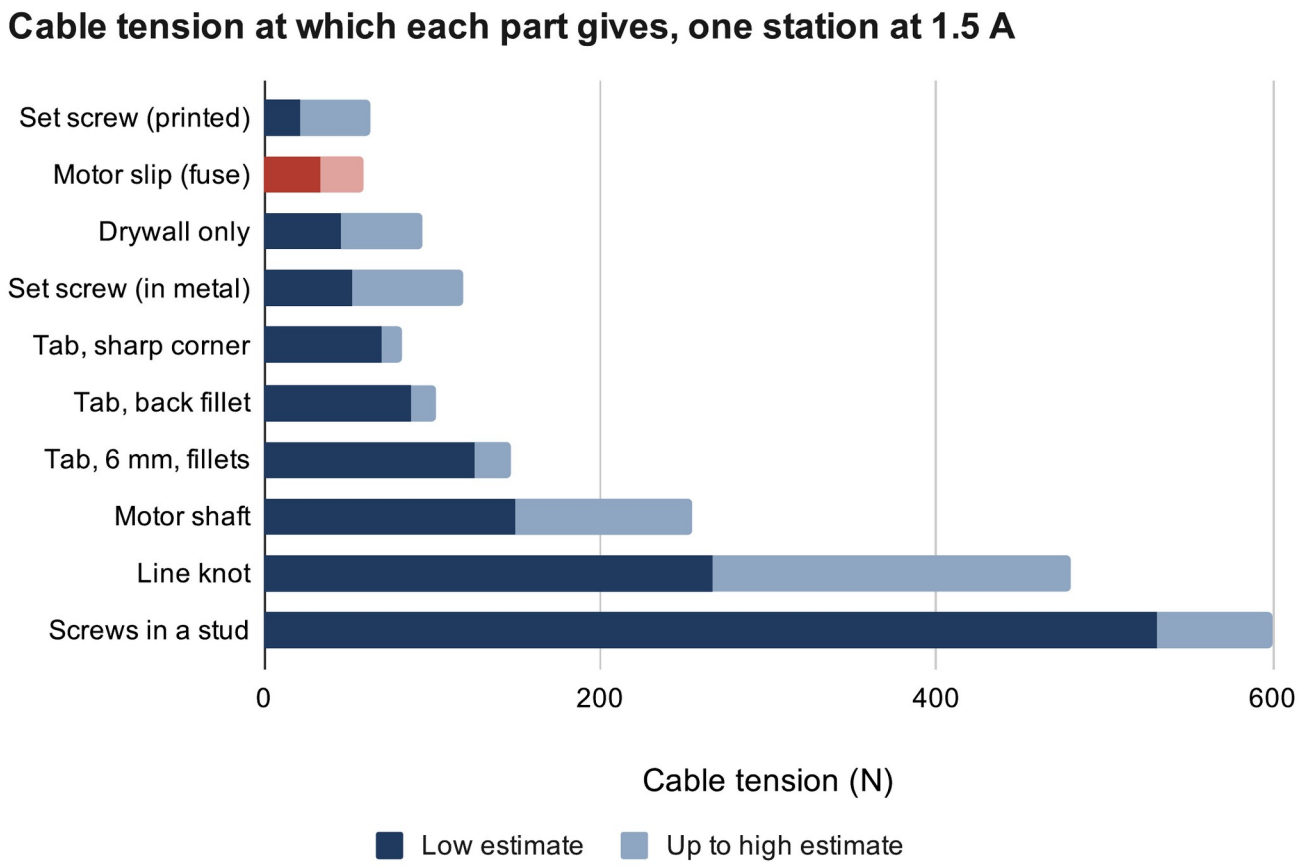
Motor slip force (the fuse)

$F = \tau_h / r = 0.36\ \text{N·m} / 0.0075\ \text{m} = \mathbf{48\ N}$

$F_{max} = 48\ \text{N} \times 1.11\ (\text{motor}) \times 1.11\ (\text{cable angle}) = \mathbf{59\ N}$

The drive slips before the wall does: a designed fuse

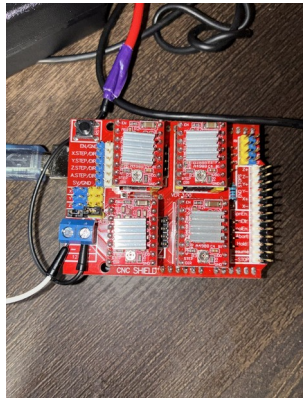
CALCULATED

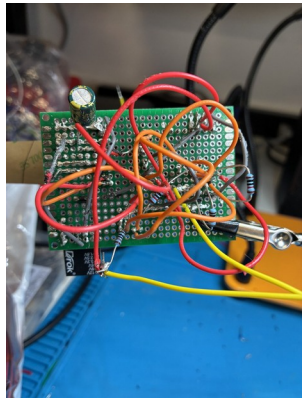


- Four cables on three axes can pull against each other; a length error becomes tension, up to a snag that could tear a mount out
- The motor's rotor skips at 34–59 N of cable tension, the designed fuse: every other part gives higher, the wall screws in a stud at 9 × and the slide-in dovetails at 225 × the motor's pull
- The spool encoder sees the spool fall behind the commanded steps: a TRACKING fault, and every station holds
- The same mechanism catches an overweight pickup and a missed over-tension; the one exception as drawn, the set screw in printed plastic (22–64 N), moves into a metal insert

Calculated (calc sheet: Weakest link)

One controller per winch





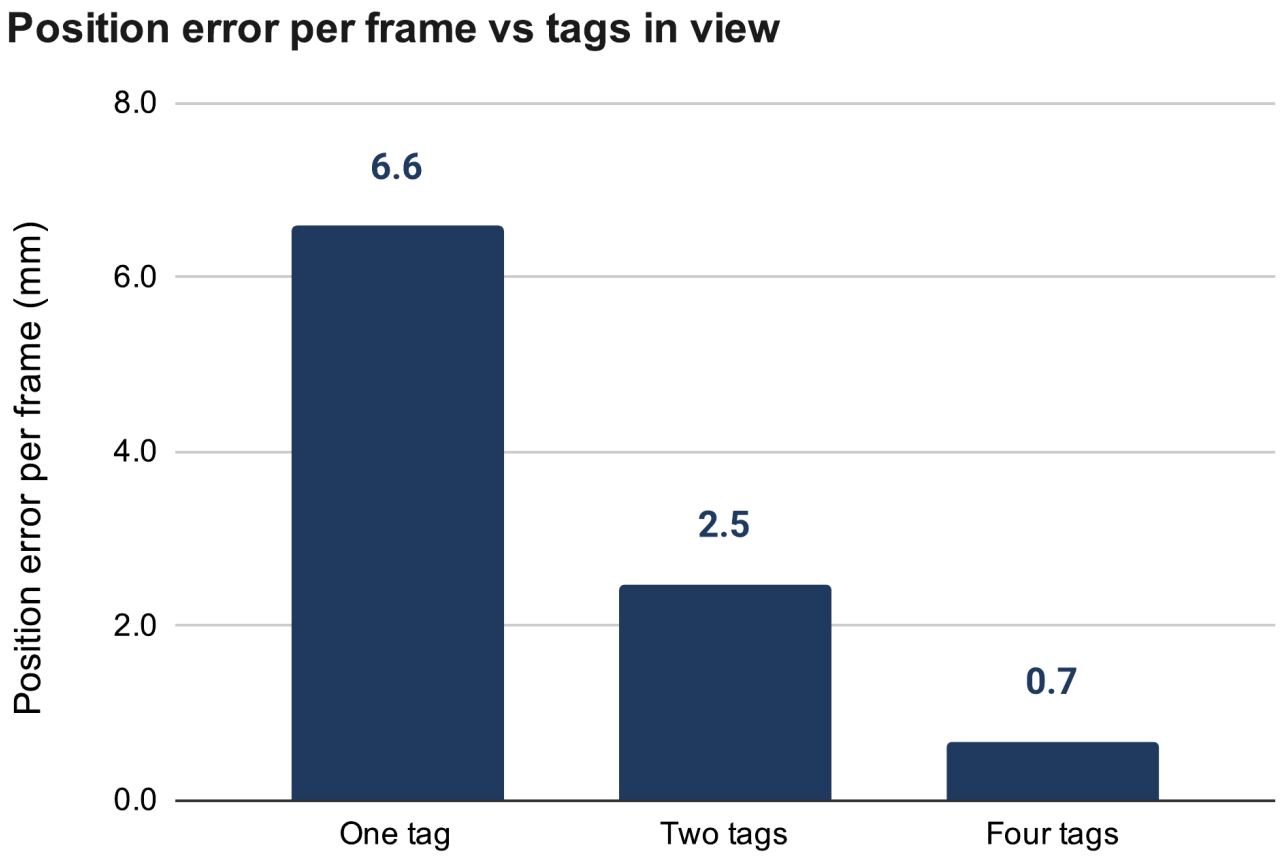
Before: one central driver board. After: a controller in each station.

	Generation 1	Final
Where is the claw?	Assumed from commanded cable lengths	Measured by the claw camera at every startup
Room geometry	Anchors from a tape measure	Anchors and cable lengths fitted, with error bars
Motor control	One central board	A controller per station, coordinated by the laptop
Finding laundry	Ceiling camera	Claw camera
Did the grab work?	Assumed	Camera checks pickup and delivery
Faults	Emergency stop only	PAUSE keeps torque; a power loss drops the claw (no brake) and nothing resumes by itself

- Generation 1's central box worked within about a day and a half; it had to go: 250 m (820 ft) of wire once each winch had sensors, 2.5 mm of skew per 50 ms, a tangle to unplug and shelve, no modularity
- Now: a controller in each station, fed by one 12 V pair: 50 m (164 ft)
- Each station is built, tested and swapped on its own
- The cost: timing moves into software

The claw camera replaces bead homing

SIMULATED

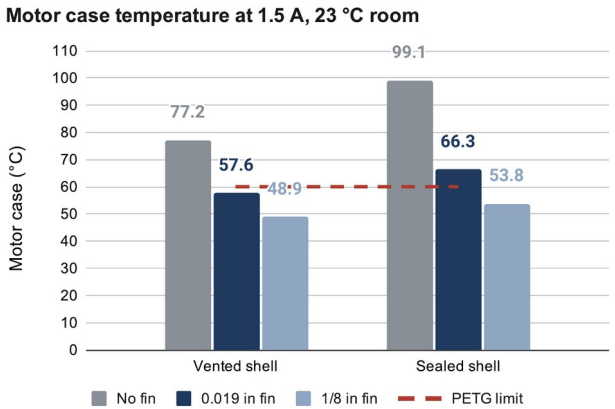


Calculated (calc sheet: Floor board)

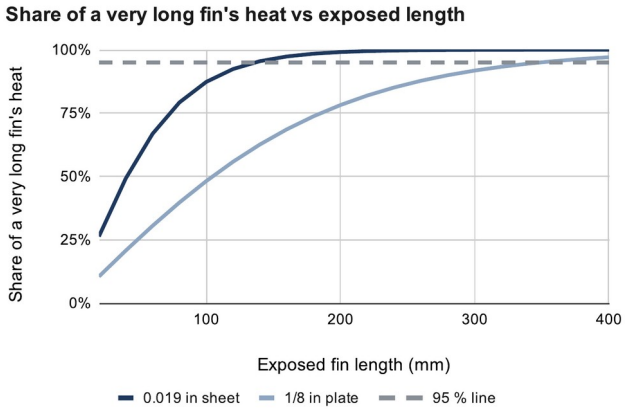
- The claw camera reads a printed floor board, and a fit finds the anchors and cable lengths: the whole room is reachable
- Camera pose per frame, best case: 6.6 mm with one tag in view, 0.7 mm with four
- After the fit, cable lengths are known to about 12–16 mm (simulated); the room opens only when the error bars allow it
- Beads stay on the cables as end stops: a bead reaching its switch stops that station

Heat: a sealed PETG box would cook its own motor mount

CALCULATED



Calculated (calc sheet: Motor temperature; assumed 1.8 Ω coil, 23 °C room)



Calculated (calc sheet: Heat fin; Incropera, ch. 3)

- 1.5 A doubles motor heat to 4.05 W; the PETG mount must stay below 60 °C
- The fin conducts heat off the motor flange and out of the shell to room air: 35–50 % of the 4.05 W
- Vented shell and fin: 56–59 °C with the 0.019 in prototype, 45–49 °C with the 1/8 in design
- Thin sheet stops gaining by 136 mm; 1/8 in plate keeps gaining to 349 mm

If the fin alone carried all 4.05 W (painted 6061, 44 mm wide)

Thickness	Mass	Shortest for ≤ 60 °C
0.019 in (prototype)	14 g	None (2.0 W max)
1/16 in	47 g	None (3.6 W max)
0.090 in	68 g	210 mm or none
1/8 in (design)	94 g	145–185 mm
3/16 in	141 g	125–140 mm
1/4 in	189 g	115–125 mm

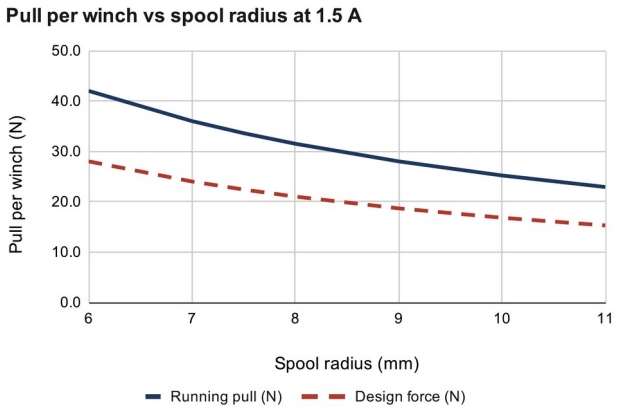
Motor heat, $P = I^2 R$

At 1.06 A: $(1.06\ \text{A})^2 \times 1.8\ \Omega = 2.0\ \text{W}$

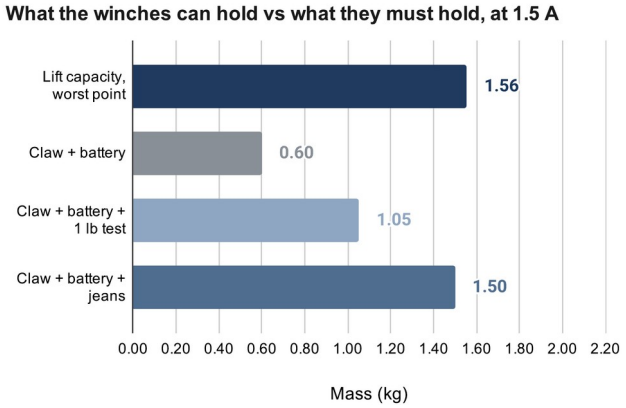
At 1.5 A: $(1.5\ \text{A})^2 \times 1.8\ \Omega = \mathbf{4.05\ W}$

Winch pull and what the four winches can lift

MEASURED CALCULATED



Calculated (calc sheet: Winch pull)



Calculated (calc sheet: Lift budget)

- Calculated pull at 1.5 A: 34 N on an empty spool, 25 N when full; running torque assumed 0.7 × holding
- Claw with battery weighed 0.60 kg (0.86 kg estimated)
- Jeans: 1.50 kg against 1.56 kg capacity, a 4 % margin, inside the uncertainty (1.1 kg at 0.5 × holding torque)
- The pull test decides: keep 1.5 A, or add a gear stage at each winch

Lift capacity

$$m_{max} = F_{design} / (k\,g) = 16.8 / (1.101 \times 9.81)$$

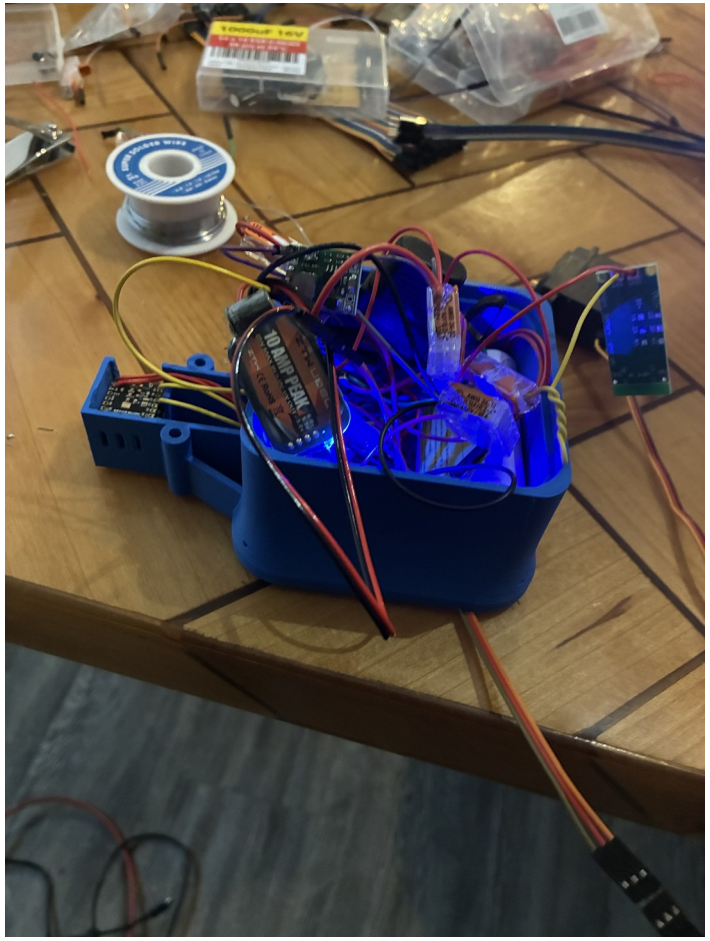
$$= \mathbf{1.56\ kg} \text{ against } 0.60 + 0.90 = \mathbf{1.50\ kg}$$

The claw: geared four-bar hands, camera bay, one 2S pack

CALCULATED



1 Claw assembled (Oct 22)



2 Electronics housing open (Oct 26)

- The hands are four-bar couplers: they stay within 1° of parallel over the 53 mm opening
- Grip 13.3–25 N per jaw against 17.7 N for jeans: short at the narrowest grab, enough above a 22 mm grab
- One 7.4 V two-cell pack: a 6 V regulator for the servo, a 5 V buck for the board, a 200k/100k divider for battery sense

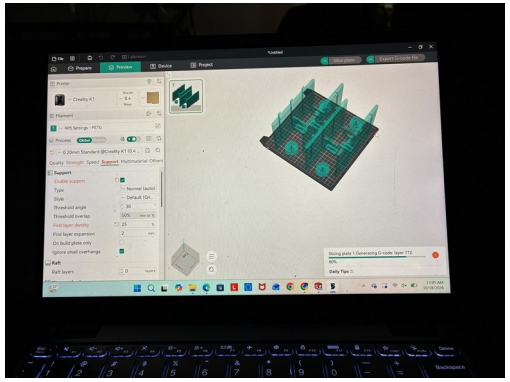
Grip force per jaw, jeans

$N \geq m\,(g + a) \,/\, (2\mu) = 0.9 \times 11.81 \,/\, 0.6 =$ **17.7 N**

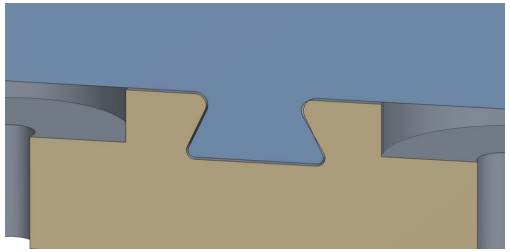
Jaws: **13.3 N** narrowest, 25 N at a 70 mm grab

Printing and machining: fits came from coupons, not guesses

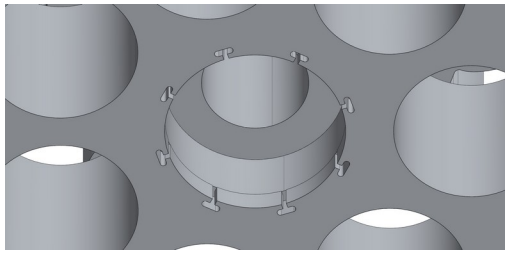
MEASURED



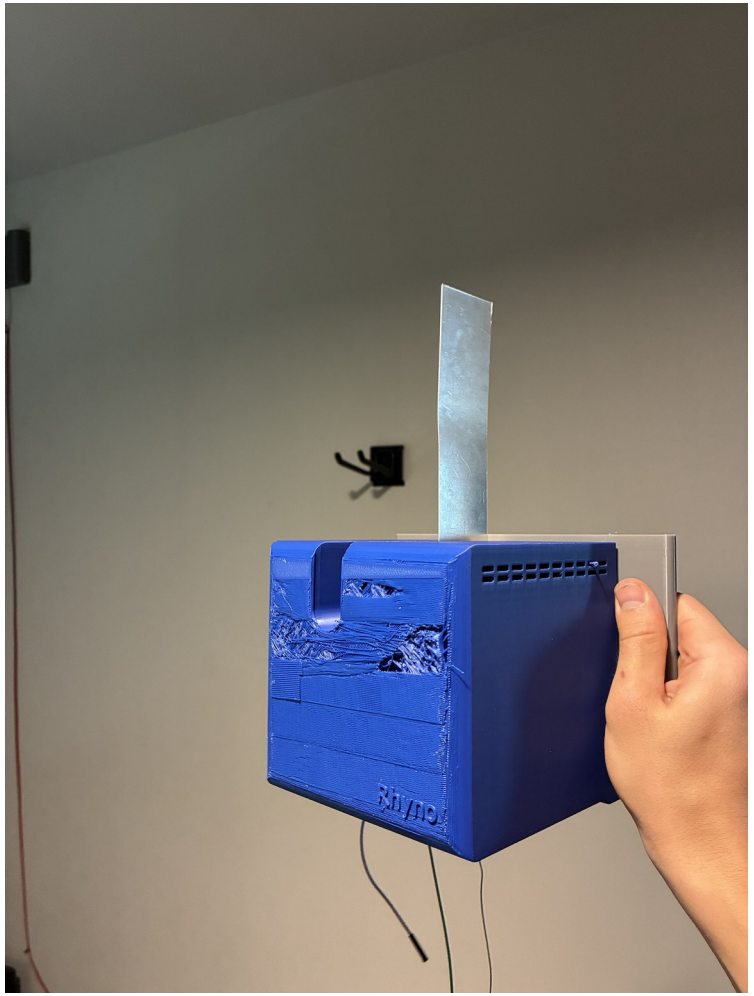
1 Slicing PETG, 0.2 mm layers



2 Slide-in rib in the dovetail



3 Gear: Ø14.000 entry lip, 8 T-slots

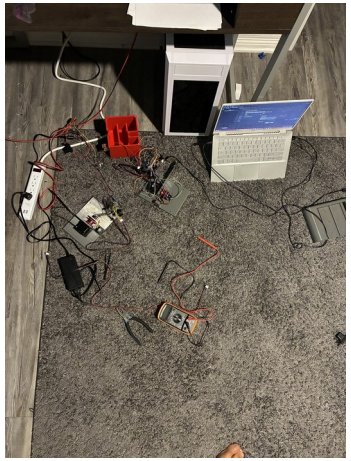


4 Blue shell and fin; the failed layer near the top was kept

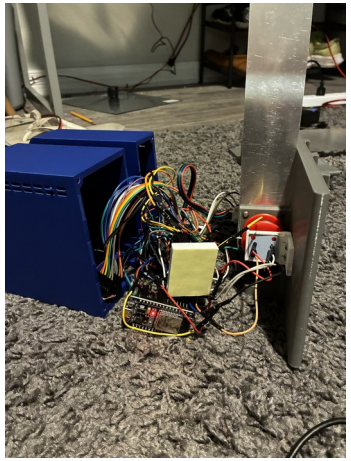
- The spool bore printed too tight (Ø5.2 on a 5 mm shaft): opened to Ø5.45 after a fit coupon
- Fin holes drilled in one setup: the first hole became the mill's zero, so the pattern carries ±0.1 mm, not the sheared edges' ±0.5
- One shell kept a failed layer: it is a shell, not a load path
- Printed holes come out small, so every printed fit carries an allowance; the drilled fin needs none

Station electronics on the bench: every check before the motor turns

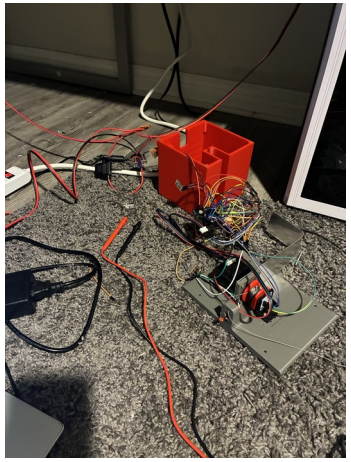
MEASURED



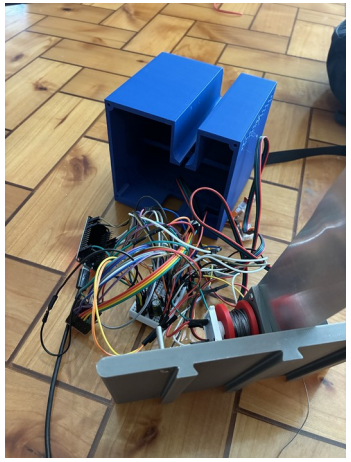
1 Bench: two stations, laptop, meter, 12 V supply



2 Dev board, driver, spool and end stop



3 Red shell, fused 12 V lead

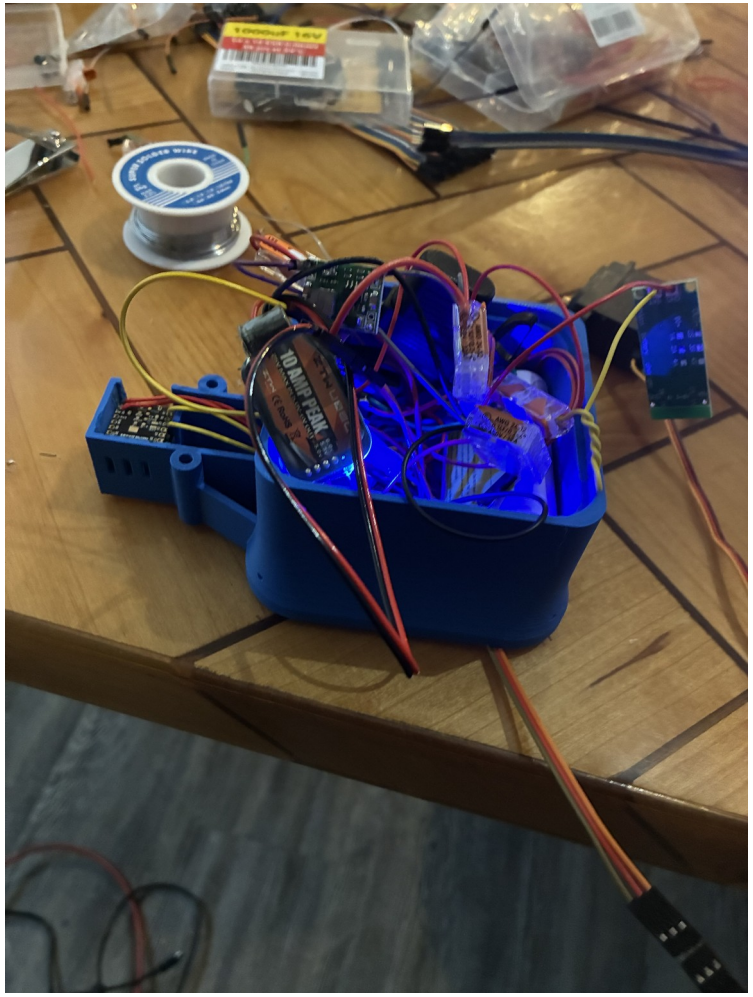


4 Second board, shell open

- Driver off at boot: a reset cannot move a motor
- Homing switch LOW at rest, HIGH pressed or with a broken wire; the encoder answers at its address with the magnet detected
- One spool turn within 60 encoder counts: PASS on both boards. The spool encoder (magnetic, on the spool shaft, Ø4 magnet) has two jobs: confirm every commanded step, and detect slip as the tension sensor
- A short found by the board's LED going out, traced to the switch pull-up

Claw electronics on the bench: camera, battery, servo

MEASURED



1 Housing open: regulators, battery, camera board

[Jasper: screenshot of the live view on the phone]

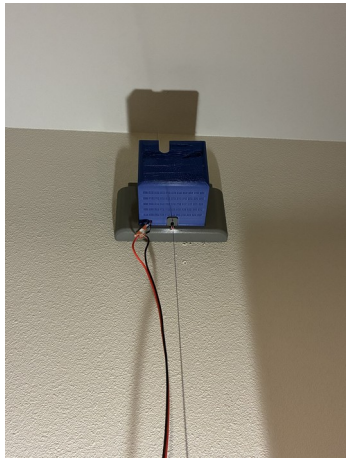
2 Live view from the claw camera

- Live video from the claw camera on the phone
- Battery 7.2 V read through the divider, matching the meter
- The servo only moved once it had its own hardware timer; the camera owned the first one
- Still open: servo end points unmeasured; camera not yet calibrated

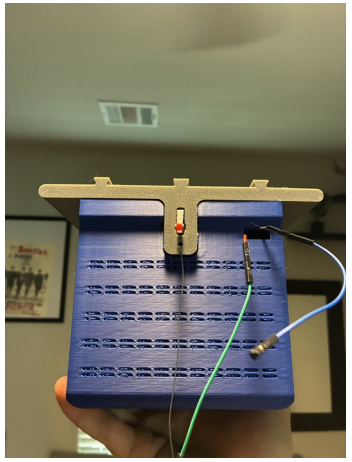
On the wall



1 Station mounted high on the TV wall, 12 V leads down



2 From below: the line leaves the switch tab straight down

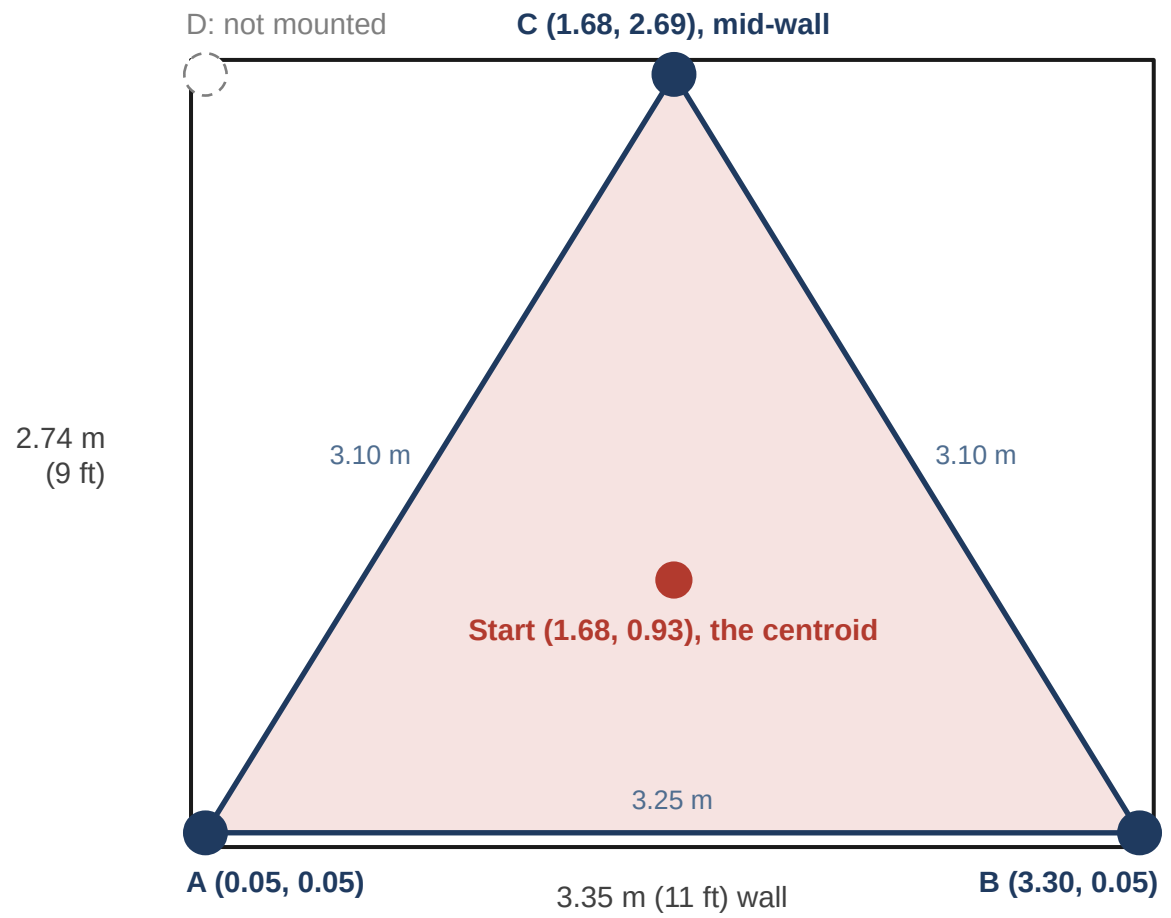


3 Shell front: vents, fin tab

- First station in a stud on the TV wall: 12 V down the wall, the fin out through the slot in the top
- Boxed stations boot and join Wi-Fi on their own; firmware updates arrive over Wi-Fi, so a boxed station never needs a USB cable again
- The motor stays off until the phone arms it; a station accepts an update only while disarmed

Three stations first: the triangle, the tilt, and no spare cable

SIMULATED



Top view to scale; anchors at one height, 2.59 m; shaded: where three cables hold the claw

- A and B in two corners of one wall, C mid-wall opposite: a 3.25 / 3.10 / 3.10 m triangle, nearly equilateral
- Hanging from three of the four corner holes tilts the claw 5.5° at the start point; the lift aborts at 8°
- Grips (5° limit) would be refused over 78 % of the floor; a three-hole housing at 120° brings the start tilt to 0.8°
- No spare cable: a station fault drops the claw, so the floor under it stays clear and padded

What hurts now, and the next iteration

CAD | PLANNED

WINCH PULL

A 3:1 printed gear stage

Now: a direct-drive spool on the motor shaft, 34 N on an empty spool at 1.5 A

NEXT ITERATION

- 18T pinion, 54T gear, module 1, printed PETG pinion: designed and CAD-verified, pinion printed, not fitted
- Lift 1.98–2.80 kg at 1.0 A with half the heat; the slip fuse moves to about 130 N, so that station is sized for 200 N

CAD, not fitted

SLIDE-IN FIT

A 1° draft on the dovetail

Now: the slide-in binds on insertion; its ribs run parallel on 0.175 mm a side

NEXT ITERATION

- A 1° draft on the dovetail and the interlocking supports, thicker at the top and thinning toward the bottom
- The station self-locates as it slides in and does not bind

PLANNED

WALL MOUNT

A lighter back mount

Now: the mount holds about 200 × the motor's pull (dovetails 225 ×, screw heads 93 ×): overbuilt, 597 cm³ solid

NEXT ITERATION

- Thinner walls and ribs, about 200 cm³: less print time and filament
- For anyone building it from the open-source files

PLANNED

[Jasper: photos or CAD of the next iteration]

Next-iteration parts: gear stage, drafted slide-in, lighter mount

Results against targets, and what is still planned

MEASURED | CALCULATED | PLANNED

Target	Goal	Current	Evidence	Status
Lift capacity	1.50 kg with jeans	1.56 kg; 1.1 kg at 0.5 × holding torque; 1.98–2.80 kg with the 3:1 stage (next iteration)	Calculated	Marginal; pull test
Grip for jeans	17.7 N per jaw	13.3–25 N; a narrow grab is a jaw gap under 22 mm at the cloth (6 V), where the lever arm is longest	Calculated	**Short at narrow grabs**
Claw mass	0.45 kg	0.60 kg with battery	Weighed	**Not met**
Motor temperature	Below 60 °C	56–59 °C prototype fin	Calculated	Marginal; hold test
Weakest link	Motor slips first	Set screw may slip first; tab at 1.2 ×	Calculated	**Two fixes**
Working area	Floor, 0.2 m wall margin	All but 0.2 m at the walls	Calculated	Met in design
Wiring	One cable per station	One 12 V pair; 0.51 V drop	Built	Met (bench PASS)
Station bench test	Switch, encoder, driver, one turn	PASS on both boards	Measured	Met
Claw bench test	Camera, battery, servo	Live view; 7.2 V; servo moves	Measured	Met
Accuracy	≤ 2 mm cable mismatch	40 ms skew budget gives 2 mm	Not measured	To be tested
Obstacle map in the app	Drawn shapes kept out of every path	Built; two findings being fixed	Simulated	Planned
First three-station lift	Self-check, calibration, short move	Not yet run	None	Planned
Cost	Hobby budget	About $640 ordered; $220–260 in the robot (3–4 stations), prices partly estimated	Receipts	Met

What I would do first next time, and what happens next

Good decision	Result
One controller per winch	Five times less wire; each station tested alone
Camera calibration instead of bead homing	Whole room reachable; caught on paper before building
A mass budget per part	The claw was weighed; the next iteration's gear stage sized to the lift margin

Poor decision	Result
Mass not tracked through four claw versions	v6 reached 4.6 times its budget in CAD; the claw is 0.15 kg over
Prototype fin cut from scrap sheet	Only 1–4 K of thermal margin
Build and test week ending on presentation day	System results arrive at the last minute

- Half the time: pull-test one winch and weigh a mock claw in week 1; kill the central board, bead homing and the fingers on paper; prove one station first

Next

- Flash and commission the three stations on the wall
- Pull test at 1.5 A; enter the measured limit in the planner
- First lift: self-check, calibration, a short move with 1 lb
- Then the robot scans its own room: height map and obstacle proposals in the app
- A brake or non-backdrivable stage, so a power loss cannot drop the claw
- A pole-clamp back for the wall mount (a gym-pole clamp), so a station mounts to a pole, not only a wall
- Open-source build on GitHub, so others can build it